

Chapter 27:

Quantum Phase Estimation

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General Controlled Unitary Gate

- The single-qubit arbitrary unitary gate, $U_{\theta,\phi,\lambda}$ describes how a single qubit rotates on the Bloch sphere.
- We discarded the global phase information because for a single qubit, the global phase has no physical significance.
- The rotation on a Bloch sphere can be described by the $U_{\theta,\phi,\lambda}$ defined as following:

$$U_{\theta,\phi,\lambda} = \begin{pmatrix} \cos \frac{\theta}{2} & -e^{i\lambda} \sin \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} & e^{i(\lambda+\phi)} \cos \frac{\theta}{2} \end{pmatrix}$$

This gate represents all possible one-qubit gates as long as we do not care about the global phase of the state. But it is a problem when multiple qubits are considered.

- When more qubits are involved, while the global phase of the whole system does not matter, the “global phases” of the individual qubits matter.
- The global phases of the individual qubits determine the relative phase between different qubits and can result in constructive or destructive interferences that are important mechanisms in quantum computing.

General Controlled Unitary Gate

- Therefore, *when we are not allowed to neglect the global phase of a single qubit*. We can extend the concept, so it will be able to represent any unitary rotation of a qubit by adding a phase factor, γ , as the following:

$$U_{\theta, \phi, \lambda, \gamma} = e^{i\gamma} \begin{pmatrix} \cos \frac{\theta}{2} & -e^{i\lambda} \sin \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} & e^{i(\lambda+\phi)} \cos \frac{\theta}{2} \end{pmatrix}$$

The θ , ϕ , and λ still have the same meaning as in $U_{\theta, \phi, \lambda}$.

- We can see that when we only have $U_{\theta, \phi, \lambda}$, for some matrices, we are missing the $e^{i\gamma}$. This causes an error when it is applied to a vector by omitting $e^{i\gamma}$. But for a single-qubit case, missing a phase does not matter.

General Controlled Unitary Gate

- For example, represent the gate $\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$ in the form of $\mathbf{U}_{\theta, \phi, \lambda, \gamma}$.

$$\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = i \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = e^{i\pi/2} \begin{pmatrix} \cos \frac{0}{2} & -e^{i\pi} \sin \frac{0}{2} \\ e^{i0} \sin \frac{0}{2} & e^{i(\pi+0)} \cos \frac{0}{2} \end{pmatrix}$$

- Therefore, $\mathbf{U}_{0, 0, \pi, \pi/2}$ is the correct representation. When we apply this gate to $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$, we get

$$\mathbf{U}_{0, 0, \pi, \pi/2} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = i \begin{pmatrix} \alpha \\ -\beta \end{pmatrix}$$

- If we use $\mathbf{U}_{0, 0, \pi}$ instead, a phase factor of $e^{i\gamma} = e^{i\pi/2} = i$ is missing. When we apply $\mathbf{U}_{0, 0, \pi}$ to $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$, it becomes

$$\mathbf{U}_{0, 0, \pi} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \alpha \\ -\beta \end{pmatrix}$$

The rotated state vectors are the same except differ by the phase factor, i .

General Controlled Unitary Gate

- To construct a general version of a controlled unitary gate. Since two qubits are involved, we cannot ignore the global phase of each qubit anymore.
- Therefore, the controlled unitary gate must use all four parameters, and its symbol is $\mathcal{C} - U_{\theta, \phi, \lambda, \gamma}$.

- It has this matrix:

$$\mathcal{C} - U_{\theta, \phi, \lambda, \gamma} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & e^{i\gamma} \cos \frac{\theta}{2} & -e^{i(\lambda+\gamma)} \sin \frac{\theta}{2} \\ 0 & 0 & e^{i(\phi+\gamma)} \sin \frac{\theta}{2} & e^{i(\lambda+\phi+\gamma)} \cos \frac{\theta}{2} \end{pmatrix}$$

- Here we assume the MSB is the control qubit and the LSB is the target qubit. We can find this matrix by using the definition. A $\mathcal{C} - U$ gate means that if the control qubit in the basis vector is 0, it does nothing to the target qubit, and if the control qubit is 1, it will apply the unitary gate. Therefore, the definition can be written as:

$$\begin{aligned} \mathcal{C} - U_{\theta, \phi, \lambda, \gamma} |00\rangle &= I|0\rangle \otimes I|0\rangle = |00\rangle \\ \mathcal{C} - U_{\theta, \phi, \lambda, \gamma} |01\rangle &= I|0\rangle \otimes I|1\rangle = |01\rangle \\ \mathcal{C} - U_{\theta, \phi, \lambda, \gamma} |10\rangle &= I|1\rangle \otimes U_{\theta, \phi, \lambda, \gamma} |0\rangle \\ \mathcal{C} - U_{\theta, \phi, \lambda, \gamma} |11\rangle &= I|1\rangle \otimes U_{\theta, \phi, \lambda, \gamma} |1\rangle \end{aligned}$$

General Controlled Unitary Gate

$$\begin{aligned}
\mathcal{C} - U_{\theta, \phi, \lambda, \gamma} |00\rangle &= I|0\rangle \otimes I|0\rangle = |00\rangle \\
\mathcal{C} - U_{\theta, \phi, \lambda, \gamma} |01\rangle &= I|0\rangle \otimes I|1\rangle = |01\rangle \\
\mathcal{C} - U_{\theta, \phi, \lambda, \gamma} |10\rangle &= I|1\rangle \otimes U_{\theta, \phi, \lambda, \gamma} |0\rangle \\
&= |1\rangle \left(e^{i\gamma} \cos \frac{\theta}{2} |0\rangle + e^{i(\phi+\gamma)} \sin \frac{\theta}{2} |1\rangle \right) \\
&= e^{i\gamma} \cos \frac{\theta}{2} |10\rangle + e^{i(\phi+\gamma)} \sin \frac{\theta}{2} |11\rangle \\
\mathcal{C} - U_{\theta, \phi, \lambda, \gamma} |11\rangle &= I|1\rangle \otimes U_{\theta, \phi, \lambda, \gamma} |1\rangle \\
&= |1\rangle \left(-e^{i(\lambda+\gamma)} \sin \frac{\theta}{2} |0\rangle + e^{i(\lambda+\phi+\gamma)} \cos \frac{\theta}{2} |1\rangle \right) \\
&= -e^{i(\lambda+\gamma)} \sin \frac{\theta}{2} |10\rangle + e^{i(\lambda+\phi+\gamma)} \cos \frac{\theta}{2} |11\rangle
\end{aligned}$$

- E.g. find the element ($e_{2,2}$) of the second row and second column (counting from 0-th) of $\mathcal{C} - U_{\theta, \phi, \lambda, \gamma}$.

The element corresponds to $\langle 10|$ (row) and $|10\rangle$ (column). Using the above equation:

$$e_{2,2} = \langle 10|\mathcal{C} - U_{\theta, \phi, \lambda, \gamma}|10\rangle = e^{i\gamma} \cos \frac{\theta}{2}$$

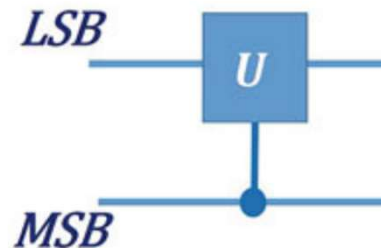
- When we apply $\langle 10|$ and we use the orthonormal property of the basis vectors that $\langle 10|10\rangle = 1$ and $\langle 10|11\rangle = 0$. This is the same element $e_{2,2}$ as in the matrix $\mathcal{C} - U_{\theta, \phi, \lambda, \gamma}$

General Controlled Unitary Gate

- Finally, we may also construct the matrix $C = U_{\theta, \phi, \lambda, \gamma}$ using this equation:

$$C = U_{\theta, \phi, \lambda, \gamma} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U_{\theta, \phi, \lambda, \gamma}$$

- It is easy to see the result if we recognize that $|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $|1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. We then perform the tensor products accordingly.
- This equation says that we should perform identity operation to the target qubit if the control qubit is 0 and apply the unitary gate to the target qubit if the control qubit is 1.
- Pic. 45 shows the gate symbol.



Pic. 45 Symbol of a $C = U_{\theta, \phi, \lambda, \gamma}$ gate with the MSB as the control qubit

Quantum Phase Estimation

- The purpose of the **Quantum Phase Estimation (QPE)** algorithm is to estimate the phase, $2\pi\psi$, of the eigenvalue of a unitary $m \times m$ matrix, U .
- Since it is a unitary matrix, its eigenvalue must have a magnitude of one. This is because we can always transform the matrix into a basis formed by its eigenvectors (as the basis states), and its diagonal elements are just its eigenvalues.
- And since the columns of the unitary matrix must be normalized, then the eigenvalues must have a magnitude of one and be in the form of $e^{i2\pi\psi}$. We choose to write its phase as $2\pi\psi$ instead of a single variable is just for convenience in the derivation.
- Let us look at a QPE example for a 2×2 matrix. Assume we are given a phase shift gate with $\phi = \pi$, $U_{PS,\pi}$ which is just a Z-gate, Z .

$$U_{PS,\pi} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} e^{i2\pi 0} & 0 \\ 0 & e^{i2\pi \frac{1}{2}} \end{pmatrix}$$

We see that it is already in its diagonal form, and therefore, the elements are its eigenvalues, 1 and -1 , which have phases of 0 and π , respectively. We express them in the form of $e^{i2\pi\psi}$, $\psi_0 = 0$ and $\psi_1 = 1/2$.

Quantum Phase Estimation

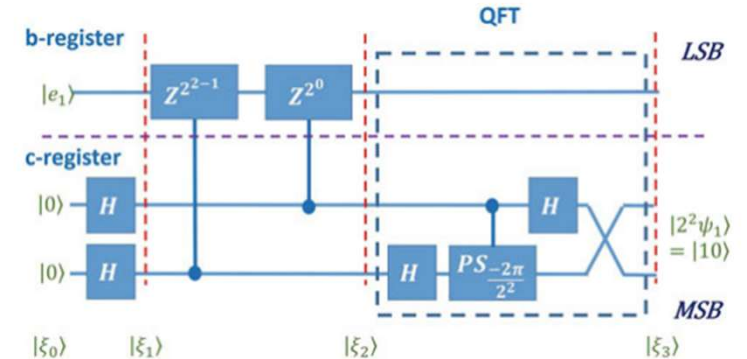
- Each of the eigenvalues corresponds to an eigenvector, and they are $|e_0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|e_1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, respectively.
- We can prove that they are the correct eigenvectors by showing, by definition,

$$U_{PS,\pi}|e_0\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1|e_0\rangle$$

$$U_{PS,\pi}|e_1\rangle = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -1|e_1\rangle$$

The goal of the QPE is to find ψ_0 and ψ_1 for the given $U_{PS,\pi}$.

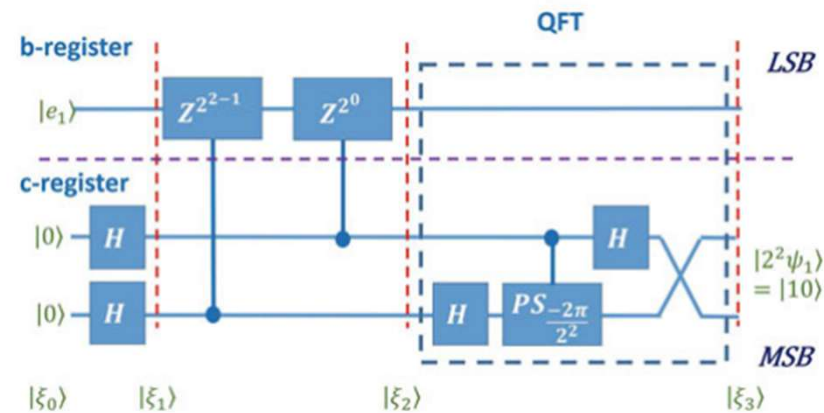
- Pic 46. shows the quantum circuit for finding the phase of the Z-gate. To be exact, it finds ψ instead of $2\pi\psi$.
- There are two groups of inputs. The less significant qubits (called the b-register) take the eigenvector of the Z-gate. Since the Z-gate is a 2×2 matrix, we only need 1-qubit to represent its eigenvector.
- In the example, we assume the input is $|e_1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. The other group of inputs (called the c-register) contains the more significant qubits. In this example, we use two qubits.



Pic. 46 The QPE circuit for finding the phase of the second eigenvalue of the Z-gate ($U_{PS,\pi}$). Note that the bottom is the MSB

Quantum Phase Estimation

- These qubits will be used to store a value related to ψ , which is $2^n\psi$, where n is the number of the qubits in the c-register. In this case, it is $2^n\psi = 4\psi$.
- The inputs to the c-register are the ground state $|0\rangle_2$. This is a $2+1 = 3$ -qubit circuit. Therefore, it starts with $|\xi_0\rangle = |0\rangle|0\rangle|e_1\rangle$.



Pic. 46 The QPE circuit for finding the phase of the second eigenvalue of the Z -gate (UPS, π). Note that the bottom is the MSB

- The circuit begins with a two-dimensional Hadamard gate ($H \otimes H$) to create a superposition on the c-register.

Quantum Phase Estimation

$$\begin{aligned}
 |\xi_0\rangle &= \mathbf{H} \otimes \mathbf{H} \otimes \mathbf{I} (|0\rangle|0\rangle|e_1\rangle) \\
 &= \mathbf{H}|0\rangle \otimes \mathbf{H}|0\rangle \otimes \mathbf{I}|e_1\rangle \\
 &= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|e_1\rangle \\
 &= \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)|e_1\rangle \\
 &= \frac{1}{2}(|00\rangle|e_1\rangle + |01\rangle|e_1\rangle + |10\rangle|e_1\rangle + |11\rangle|e_1\rangle)
 \end{aligned}$$

- After that, controlled gates are used. The type of controlled gate used is related to the Z-gate, of which we are finding the eigenvalue phase. If we were to find that of another gate/unitary matrix (e.g. the NOT gate), we need to use the controlled gate related to that new gate (e.g. CNOT gate).
- It is related to the Z-gate because, besides controlled-Z-gate, it also uses controlled- Z^k gates, where $k = 2^l$, with l running from 0 to $n - 1$, n is the number of qubits in the c-register.
- A controlled- Z^k gate means that, for a basis state, if the control qubit is 0, it does nothing to the target qubit. If the control qubit is 1, it will apply Z^k gate to the target qubit.
- Each qubit in the c-register acts as the control bit of one of the controlled gates. A Z^k gate is equivalent to applying Z-gate k times. In this case, l runs from 0 to 1 and thus k can be 1 or 2.

Quantum Phase Estimation

- What happens to the b-register qubit when the controlled- Z^k gates are applied? Note that the b-register has the eigenvector of $\mathbf{Z}$.
- Therefore, the operation will give the eigenvalue of Z because of $U_{PS,\pi}|e_1\rangle = \mathbf{Z}|e_1\rangle = e^{i2\pi\frac{1}{2}}|e_1\rangle = -1|e_1\rangle$.
- And if it is $\mathbf{Z}^2$, it becomes $(e^{i2\pi\frac{1}{2}})^2|e_1\rangle = |e_1\rangle$ as $\mathbf{Z}$ is applied twice. However, this only happens if the corresponding control qubit is 1. Therefore, the n controlled gates produce the following results:

$$\begin{aligned}
 |\xi_2\rangle &= \frac{1}{2}(|00\rangle|e_1\rangle + |01\rangle\mathbf{Z}|e_1\rangle + |10\rangle\mathbf{Z}^2|e_1\rangle + |11\rangle\mathbf{Z}^2\mathbf{Z}|e_1\rangle) \\
 &= \frac{1}{2}(|00\rangle\mathbf{Z}^0\mathbf{Z}^0|e_1\rangle + |01\rangle\mathbf{Z}^0\mathbf{Z}^1|e_1\rangle + |10\rangle\mathbf{Z}^2\mathbf{Z}^0|e_1\rangle + |11\rangle\mathbf{Z}^2\mathbf{Z}^1|e_1\rangle) \\
 &= \frac{1}{2}(|00\rangle\mathbf{Z}^{0+0}|e_1\rangle + |01\rangle\mathbf{Z}^{0+1}|e_1\rangle + |10\rangle\mathbf{Z}^{2+0}|e_1\rangle + |11\rangle\mathbf{Z}^{2+1}|e_1\rangle) \\
 &= \frac{1}{2}(|0\rangle\mathbf{Z}^0|e_1\rangle + |1\rangle\mathbf{Z}^1|e_1\rangle + |2\rangle\mathbf{Z}^2|e_1\rangle + |3\rangle\mathbf{Z}^3|e_1\rangle) \\
 &= \frac{1}{2} \sum_{j=0}^3 |j\rangle\mathbf{Z}^j|e_1\rangle
 \end{aligned}$$

- In the first line, $\mathbf{Z}$ is applied to the b-register if qubit 0 of the c-register is one, and $\mathbf{Z}^2$ is applied to the b-register if qubit 1 of the c-register is one. This is based on the definition of the controlled gates.
- In the second line, we also add $\mathbf{Z}^0$ that is just an identity gate when the control qubits are 0. In the third line, we group the $\mathbf{Z}$'s so we can see the sum of their exponents for each basis vector. And in the fourth line, we write the basis vectors in decimal.
- So, the action of all controlled gates is equivalent to applying $\mathbf{Z}^j$ gate on the b-register if the c-register is $|j\rangle$.²⁰¹

Quantum Phase Estimation

- And since $|e_1\rangle$ is an eigenvector of $\mathbf{Z}^j$, $\mathbf{Z}^j |e_1\rangle = (e^{i2\pi\frac{1}{2}})^j |e_1\rangle$. So,

$$\begin{aligned} |\xi_2\rangle &= \frac{1}{2} \sum_{j=0}^3 |j\rangle \mathbf{Z}^j |e_1\rangle \\ &= \frac{1}{2} \sum_{j=0}^3 |j\rangle (e^{i2\pi\frac{1}{2}})^j |e_1\rangle \\ &= \frac{1}{2} \sum_{j=0}^3 (e^{i2\pi\frac{1}{2}})^j |j\rangle |e_1\rangle \end{aligned}$$

- The next step is to apply QFT. What really important is that we need to use the matrix for rotating the vector instead of the basis and the matrix is shown in following:

$$U_{QFT} = \frac{1}{\sqrt{N}} \begin{pmatrix} \omega^{-0 \cdot 0} & \dots & \omega^{-0 \cdot (N-1)} \\ \vdots & \ddots & \vdots \\ \omega^{-(N-1) \cdot 0} & \dots & \omega^{-(N-1) \cdot (N-1)} \end{pmatrix}$$

- The definition of QFT is repeated here,

$$U_{QFT} |j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{-jk} |k\rangle$$

Quantum Phase Estimation

- We only apply it to the c-register and keep the b-register intact. Therefore, we will apply $U_{QFT} \otimes I$ to $|\xi_2\rangle$, and we have the right-hand side equation:
- The first three lines just apply U_{QFT} to the c-register in $|\xi_2\rangle$, use the linear property of quantum mechanics, and then apply U_{QFT} .
- The last two lines use the definition of the n -th root of unity, ω .
- Finally, we group the terms. Note that the b-register is no longer useful now. *It is completely disentangled with the rest of the qubit, and it can be factorized out in the tensor product.*
- If we perform a measurement on the c-register, we will get one of the $|k\rangle$'s. However, if we rewrite the equation by shuffling the summations, we have

$$|\xi_3\rangle = \frac{1}{4} \sum_{j=0}^3 \left(\sum_{k=0}^3 (\omega)^{j(2-k)} \right) |k\rangle |e_1\rangle$$

- We see that each $|k\rangle$ has a coefficient of $\sum_{j=0}^3 (\omega)^{j(2-k)}$, and this is a summation of the n -th root of unity. This term is 0 unless $(2 - k) = 0$ due to the **constructive interference**.

$$\begin{aligned} |\xi_3\rangle &= U_{QFT} \otimes I \left(\frac{1}{2} \sum_{j=0}^3 (e^{i2\pi/4})^j |j\rangle |e_1\rangle \right) \\ &= \frac{1}{2} \sum_{j=0}^3 (e^{i2\pi/4})^j (U_{QFT} |j\rangle) \otimes I |e_1\rangle \\ &= \frac{1}{2} \sum_{j=0}^3 (e^{i2\pi/4})^j \left(\frac{1}{\sqrt{4}} \sum_{k=0}^3 (\omega)^{-kj} |k\rangle \right) |e_1\rangle \\ &= \frac{1}{4} \sum_{j=0}^3 \sum_{k=0}^3 (e^{i2\pi/4})^j \omega^{-kj} |k\rangle |e_1\rangle \\ &= \frac{1}{4} \sum_{j=0}^3 \sum_{k=0}^3 (e^{i2\pi/4})^j (e^{i2\pi/4})^{-kj} |k\rangle |e_1\rangle \\ &= \frac{1}{4} \sum_{j=0}^3 \sum_{k=0}^3 (\omega)^{j(2-k)} |k\rangle |e_1\rangle \end{aligned}$$

Quantum Phase Estimation

- Therefore, if we perform the measurement, we will only measure $|k\rangle$, if $(2 - k) = 0$. The basis state we will measure is always

$$|k\rangle = |2\rangle_{10} = |10\rangle_2$$

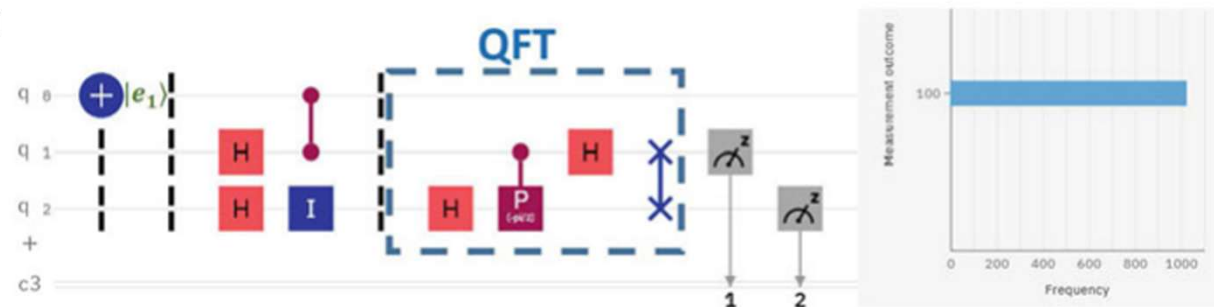
- What is the meaning of $k = 2$? Let us inspect carefully how we obtain $\omega^{j(2-k)}$ in previous equation. It came from $(e^{i2\pi\frac{1}{2}})^j \omega^{-kj}$. The “2” in $\omega^{j(2-k)}$ is obtained by multiplying $\frac{1}{2}$ by 4. And 4 is the $N = 2^2$ in the N -th root of unity. Note that $\frac{1}{2}$ is the ψ in the eigenvalue of eigenvector $|e_1\rangle$, and 4 is $2^n = 2^2$, where n is the number of qubits in the c-register. Therefore, the $k = 2$ we obtained is due to the $\psi(\frac{1}{2})$ being multiplied by $2^n(2^2 = 4)$.
- If we measure $|k\rangle$ in a n -qubit c-register, and if the input to the b-register is the eigenvector corresponding to the eigenvalue, $e^{i2\pi\psi}$, the phase is given by

$$2\pi\psi = \frac{2\pi k}{2^n}$$

$$\psi = \frac{k}{2^n}$$

Implementation on IBM-Q

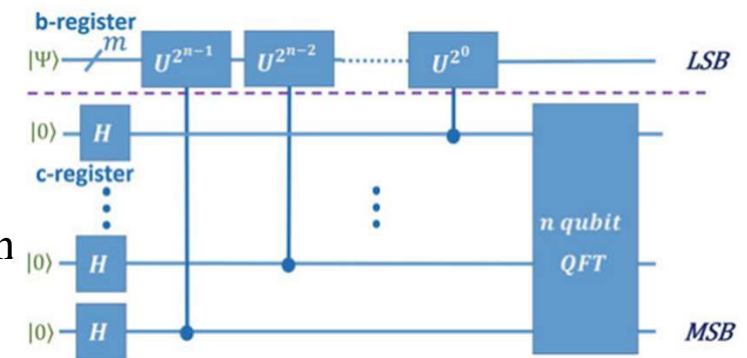
- Pic. 47 shows the implementation on IBM-Q. $|e_1\rangle$ is created by using a NOT gate.
- We can see the controlled-Z-gate that is a vertical line with two dots. There is no controlled- Z^2 gate because $Z^2 = I$. The I gate in the circuit is there just to align the circuits.
- The QFT circuit is only 2 qubits.
- You can see that the output is 100% $|100\rangle$. The first two qubits are for the c-register. Therefore, $|k\rangle = |10\rangle = |2\rangle$. That means $2\pi\psi = \frac{2\pi k}{2^n} = 2\pi 2/2^2 = \pi$, which is the phase of the corresponding eigenvalue, -1 ($e^{i2\pi\frac{1}{2}} = e^{i\pi} = -1$).
- Note that we did not use a classical register.



Pic. 47 Implementation of the QPE circuit. Note that the bottom is the MSB

General QPE Circuit

- Pic. 48 shows the general QPE circuit.
- We note that the b-register can be m -qubit that is required if we are finding the eigenvalue phase in a $m \times m$ matrix case. In that case, the controlled gate is a $m + 1$ -qubit gate (including the control bit).
- The c-register can be n qubits.
- n determines the accuracy of the phase it can estimate, larger n gives a higher resolution.
- The resolution is $1/2^n$, when $k = 1$. Usually, the ψ is not a fraction nor an integer as in the case we have discussed. Therefore, we cannot get the exact value, and thus it is an “estimation.” In summary, the QPE has $m + n$ qubits.
- The QPE seems not to be useful as we need to know the eigenvector (as input to the b-register) to find its phase. However, every vector is a linear superposition of the eigenvectors. Due to the linearity of quantum mechanics, their phases can be estimated at the same time. If we use it correctly, we can use it to solve other more important problems such as in the HHL algorithm for solving systems of linear equations.



Pic. 48 General implementation of the QPE circuit

Chapter 28: Variational Quantum Eigensolver (VQE) & Quantum Approximate Optimization Algorithm (QAOA)

Kamonluk Suksen Ph.D.

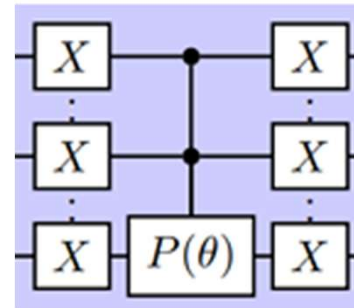
Projectors and Exponentials

- A projector operator (Π) acts on a quantum state of the Hilbert space and forces all components to zero, except those of the subspace it projects onto.

$$\Pi^2 = \Pi, \Pi|_{\text{span}(\{\psi_i\}_{i=1}^d)} = \sum_i^d |\psi_i\rangle\langle\psi_i|, \langle\psi_i|\psi_i\rangle = \delta_{ij}$$

$$e^{i\theta\Pi} = \sum_{n=0} \frac{(i\theta)^n}{n!} \Pi^n = \left(\frac{(i\theta)^0}{0!} \Pi^0 \right) + \sum_{n=1} \frac{(i\theta)^n}{n!} \Pi^n = \mathbb{I} + \left(\sum_{n=1} \frac{(i\theta)^n}{n!} \right) \Pi = \mathbb{I} + (e^{i\theta} - 1)\Pi$$

$$e^{i\theta|0\rangle\langle 0|} = \begin{pmatrix} e^{i\theta} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$



$$p(\theta) = Z^\theta = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}$$

Observables

- An observable H is a **self-adjoint operator** on the Hilbert space $\mathbb{C}^{\otimes n}$.
- Spectral theorem: $\exists$ orthonormal **basis** $\{|\psi_i\rangle\}_i$ of $\mathbb{C}^{\otimes n}$ consisting of eigenvectors of H , and all eigenvalues λ_i are **real**.
- We can write: $H = \sum_i \lambda_i |\psi_i\rangle\langle\psi_i|$
- To each energy λ_i corresponds to an **energy eigenstate**.
 - **ground state**: energy eigenstate $|V_1\rangle$ corresponding to the lowest energy
 - **first excited state, second excited state, ...**: $|V_2\rangle, |V_3\rangle, \dots$

Expectation Values

Given

- a state $|\phi\rangle$ prepared on a quantum computer using the unitary U such that $U|0\rangle = |\phi\rangle$
- an observable H we are interested to measure

Then the expectation value of H respect to the state $|\phi\rangle$ is given by

$$\langle H \rangle_{|\phi\rangle} := \langle \phi | H | \phi \rangle = \langle 0 | U^\dagger H U | 0 \rangle$$

From the spectral theorem it follows:

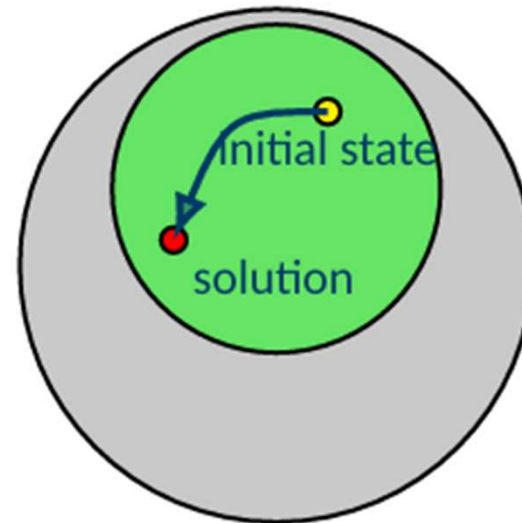
$$\langle H \rangle_{|\phi\rangle} = \langle \phi | \sum_i \lambda_i |\psi_i\rangle \langle \psi_i| \phi \rangle = \sum_i \lambda_i |\langle \psi_i | \phi \rangle|^2 = \sum_i \lambda_i |\langle \psi_i | U | 0 \rangle|^2$$

Particularly: $\langle H \rangle_{|\psi_i\rangle} = \lambda_i$

The Variational Methods

$$\langle H \rangle_{|\phi\rangle} = \sum_i \lambda_i |\langle \psi_i | \phi \rangle|^2 \geq \sum_i \lambda_{\min} |\langle \psi_i | \phi \rangle|^2 = \lambda_{\min}$$

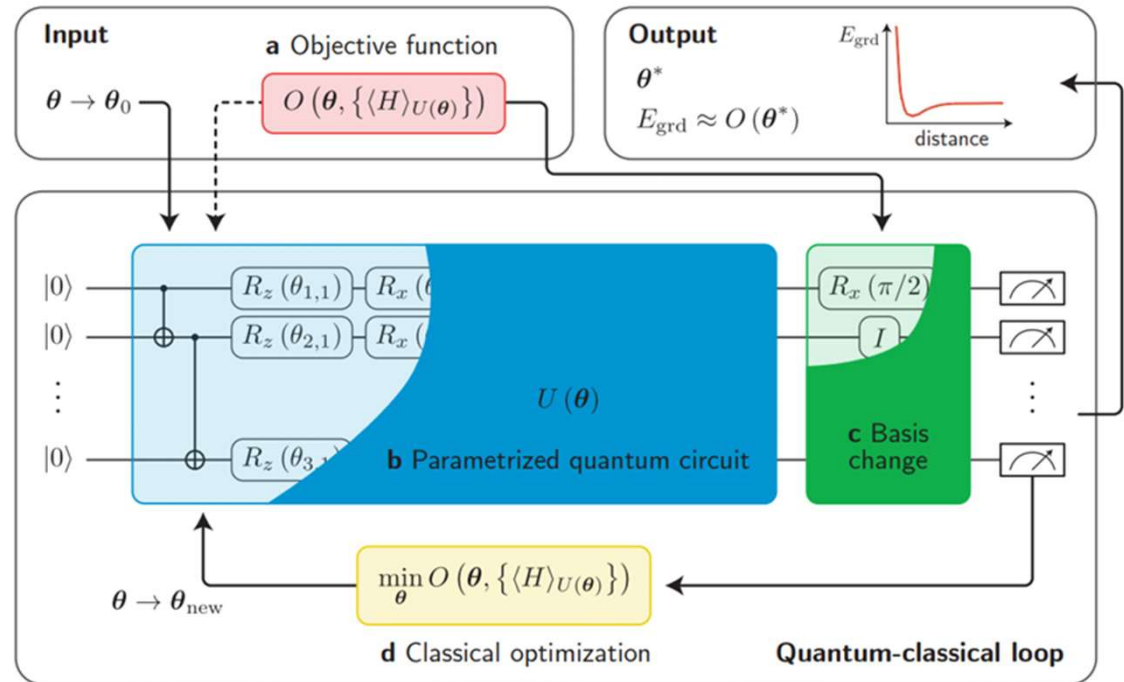
- H can encode a problem as ground state
- Prepare parameterized state $|\phi(\theta)\rangle$
- Find θ^* s.t. $\langle H \rangle_{|\psi(\theta^*)\rangle}$ minimal



The Variational Quantum Eigensolver

$$E_0 \leq \langle \psi(\theta) | H | \psi(\theta) \rangle$$

- $H = \sum_{\alpha} w_{\alpha} \vec{\sigma}_{\alpha}, \vec{\sigma}_{\alpha} \in I, X, Y, Z^{\otimes N},$
- $|\psi(\vec{\theta})\rangle = \prod_i U_{\theta_i} |0\rangle = U(\vec{\theta}) |0\rangle$
- $E_{VQE} = \min_{\vec{\theta}} \sum_{\alpha} w_{\alpha} \langle 0 | U^{\dagger}(\vec{\theta}) \vec{\sigma}_{\alpha} U(\vec{\theta}) | 0 \rangle$



Pic. 49 Quantum-Classical Optimization Loop for Variational Quantum Algorithms

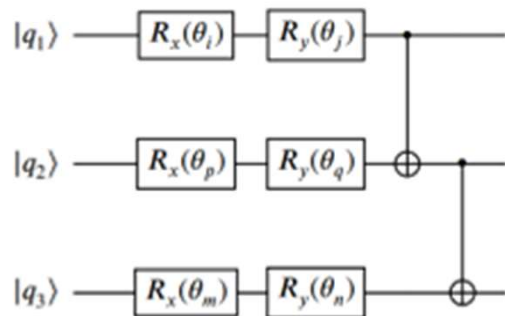
The Ansatz Problem

The right choice of ansatz is critical to obtain a solution that is close to the ground state.

- **Expressability:** Refers the range of feasible states that the ansatz can achieve.
- **Trainability:** Refers to the ability to find the best set of parameters of the ansatz respect to expectation values of the Hamiltonian in a finite amount of time.
- **Depth of the circuit:** Refers to the number of sequential operations required for the implementation, which impacts the overall runtime of the method and its resilience to noise.

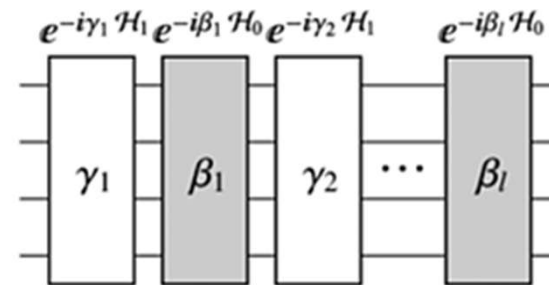
Hardware Efficient Ansatz

$$|\psi(\theta)\rangle_{HEA} = \prod_{i=1}^P U_{ent} U_{rot}(\theta_i) |0\rangle$$



Hamiltonian Variational Ansatz

$$|\psi(\theta)\rangle = \prod_{l=1}^p (\prod_j e^{i\theta_{lj}H_j}) |\psi_0\rangle, H = \sum_j H_j$$



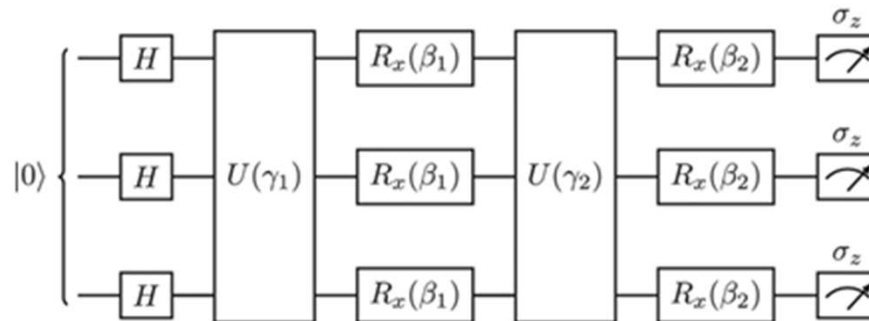
The Classical Optimizer Choice

Method	Description	Sample
Gradient Descent Based	Use the analytical property of the ansatz, the gradient of observables can be directly computed on a quantum computer.	Gradient Descent, Quantum Natural Gradient
Stochastic Gradient Based	Approximated the true gradient using random sampled data at each iteration.	SPSA, QNSPSA, Adam
Gradient-free searching	Do not rely on gradient information and instead explore the parameter space using alternative techniques as random search, evolutionary algorithms or Bayesian optimization.	COBYLA, Nelder-Mead

The Quantum Approximate Optimization Algorithm

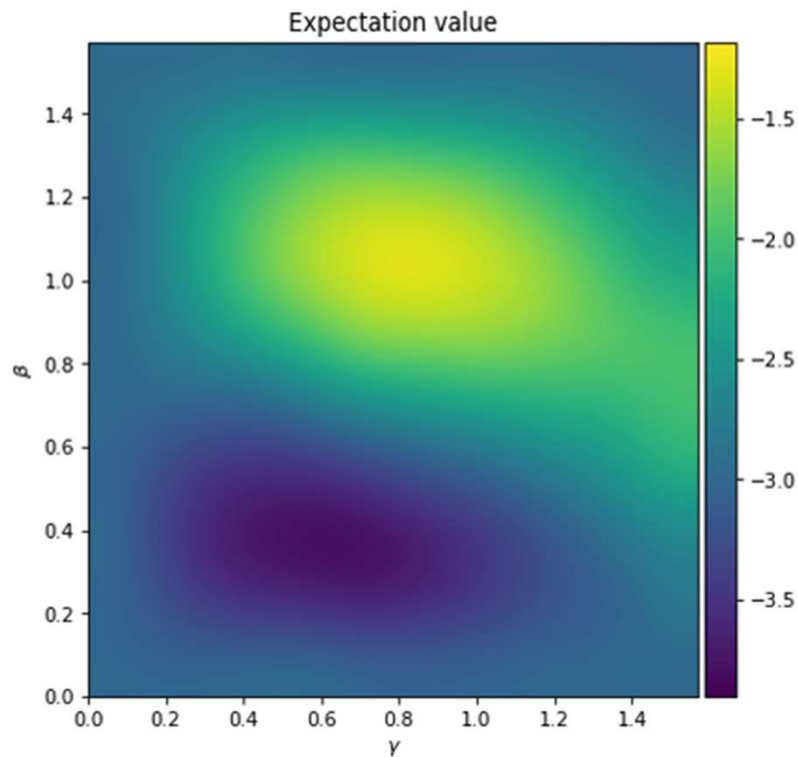
- Objective function $f: \{0, 1\}^n \rightarrow \mathbb{R}$
- Where are looking for the optimal vector $x^* = \operatorname{argmin}_{x \in \{0,1\}^n} f(x)$
- Encode each binary string into a quantum state: $z = \{0, 1\}^n \rightarrow |z\rangle$
- Encode the objective function into a problem Hamiltonian

$$H_p |z\rangle = f(z) |z\rangle, \quad \langle H_p \rangle_{|z\rangle} = f(z)$$
- The Ground state of H_p correspond to the minima of the objective function.
- $|\vec{\gamma}, \vec{\beta}\rangle = U_M(\beta_p) U_p(\gamma_p) \dots U_M(\beta_1) U_p(\gamma_1) |\phi_0\rangle$, $U_p(\gamma) = e^{-i\gamma H_p}$, $U_M = \prod_{i=1}^n R_{X_i}(\beta)$
- Find $\vec{\gamma}, \vec{\beta} \in \mathbb{R}^p$, such that $\langle \gamma, \beta | H_p | \gamma, \beta \rangle$ is minimized.

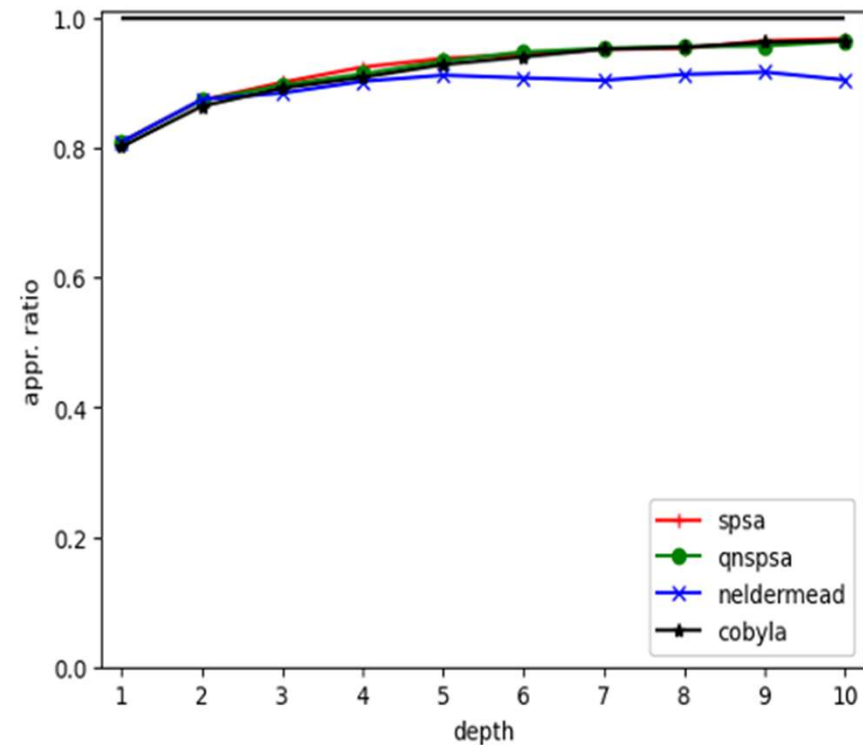


The Quantum Approximate Optimization Algorithm

Energy Landscape $p = 1$



Multiple layers QAOA performance

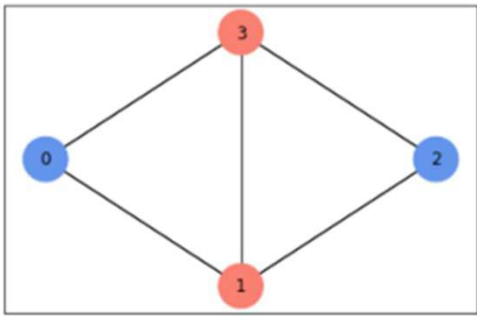


Combinatorial Optimization: Maxcut

With these binary variables, we can build a cost function for any Maxcut problem

$$H(X) = \sum_{(i,j) \in E} W_{ij} X_i (1 - X_j) \quad W_{ij} = \begin{cases} 1, & \text{if edge}(i,j) \in G & \text{connectivity} \\ 0, & \text{if edge}(i,j) \notin G & \text{(adjacency matrix)} \end{cases}$$

We claim a solution to the maxcut problem must maximize this Hamiltonian!



$$W = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

$$X = 1010 = |1\rangle_0 |0\rangle_1 |1\rangle_2 |0\rangle_3$$

$$W_{01}X_0(1 - X_1) = W_{01} \cdot 1 \cdot (1 - 0) = 1$$

$$W_{03}X_0(1 - X_3) = W_{03} \cdot 1 \cdot (1 - 0) = 1$$

$$W_{10}X_1(1 - X_0) = W_{10} \cdot 0 \cdot (1 - 1) = 0$$

$$W_{12}X_1(1 - X_2) = 0$$

$$W_{13}X_1(1 - X_3) = 0$$

$$W_{21}X_2(1 - X_1) = 1$$

$$W_{21}X_2(1 - X_1) = 1$$

$$W_{23}X_2(1 - X_3) = 1$$

$$W_{30}X_3(1 - X_0) = 0$$

$$W_{31}X_3(1 - X_1) = 0$$

$$W_{32}X_3(1 - X_2) = 0$$

$$H(X) = 4$$

Pauli Matrices

The problem of interest can be expressed in terms of Pauli matrices, I, X, Y, Z

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \vec{\sigma} = (\sigma^0, \sigma^1, \sigma^2, \sigma^3) = I, X, Y, Z$$

Such that we end up with a linear combinations of products of these Pauli matrices

$$H = \sum_i c_i P_i$$

For example:

$$H_{ex} = I_0 I_1 + X_0 X_1 + Y_0 Y_1 + Z_0 Z_1 = II + XX + YY + ZZ$$

$$\exp(-iH_{ex}t) = \exp(-i[II + XX + YY + ZZ]t)$$

Quantum state in Pauli-Z basis

Most quantum devices are in Z-basis or computational basis $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

In fact, any operations can be represented in this way:

$$\sigma_0 \equiv I \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \equiv |0\rangle\langle 0| + |1\rangle\langle 1|,$$

$$\sigma_1 \equiv X \equiv \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \equiv |1\rangle\langle 0| + |0\rangle\langle 1|,$$

$$\sigma_2 \equiv Y \equiv \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \equiv i|1\rangle\langle 0| - i|0\rangle\langle 1|,$$

$$\sigma_3 \equiv Z \equiv \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \equiv |0\rangle\langle 0| - |1\rangle\langle 1|$$

Including the Control-Not gate!

$$\text{CNOT} = (|0\rangle\langle 0|)_0 \otimes I_1 + (|1\rangle\langle 1|)_0 \otimes X_1$$

Maxcut Problem Qubit Encoding

To map the Maxcut problem to quantum circuit, we use

$$x_i \rightarrow \frac{I-Z}{2} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Then, Maxcut cost function becomes a sum of Pauli-Z and Pauli-ZZ operators (i.e., Ising Hamiltonian, or sometimes known as the QUBO problem)

$$H \rightarrow \sum_{ij} w_{ij} \frac{I_i - Z_i}{2} \left(I - \frac{I_j - Z_j}{2} \right) = \sum_i c_i Z_i + \sum_{ij} d_{ij} Z_i Z_j$$

State Preparation

State preparation (initialization) is the First step of quantum computing.

Unlike classical computation, one cannot simply prepare an arbitrary state in QC.

Often it involves careful analysis of the problem and genius engineering...

Here, we look at two possible initializations

q_0 — $|0000\rangle$

q_1 —

q_2 —

q_3 —

q_0 —  —

q_1 —  —

q_2 —  —

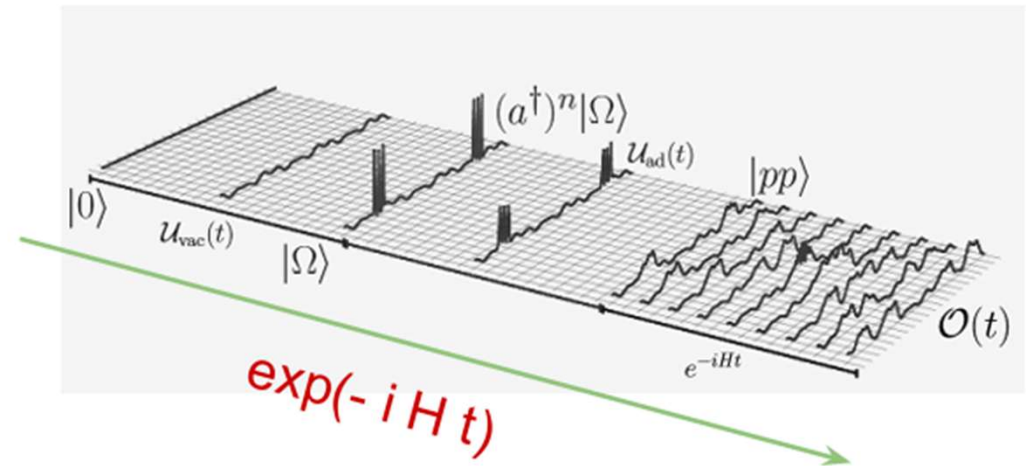
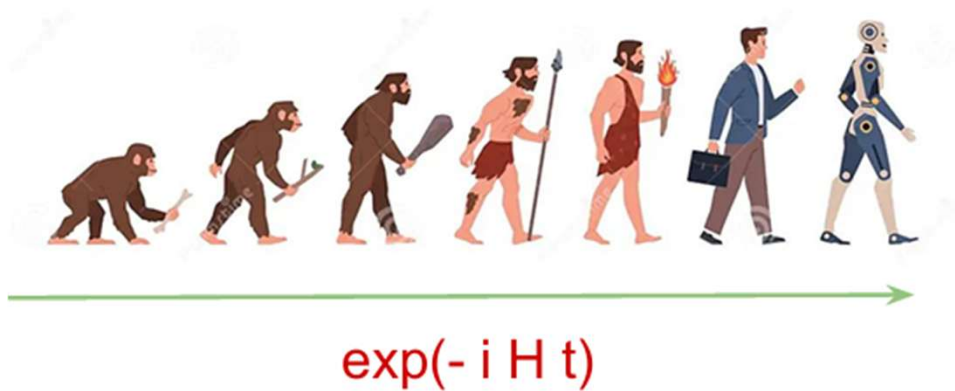
q_3 —  —

$H_0 H_1 H_2 H_3 |0000\rangle$

Hadamard Gate $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

Variational Ansatz = Unitary Quantum Evolution

Literally means evolving some quantum state by applying an operator H for a parameter t .



Single Qubit Simulation

Rotational gates take care single qubit Pauli terms (Taylor expansion)

$$e^{-iXt} = \cos(t)I - i \sin(t) X = \begin{pmatrix} \cos(t) & -i \sin(t) \\ -i \sin(t) & \cos(t) \end{pmatrix}$$

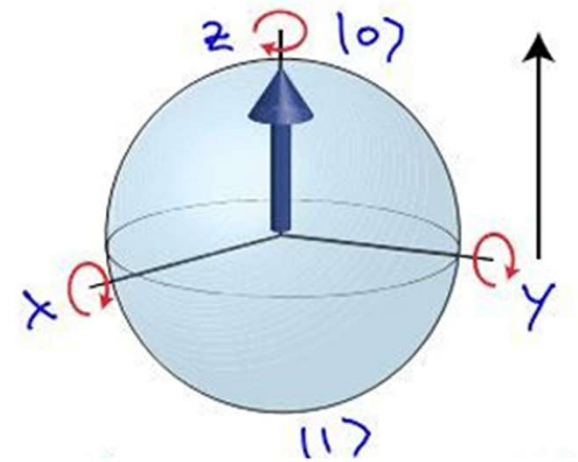
$$R_X(2t)$$

$$e^{-iYt} = \cos(t)I - i \sin(t) Y = \begin{pmatrix} \cos(t) & -\sin(t) \\ \sin(t) & \cos(t) \end{pmatrix}$$

$$R_Y(2t)$$

$$e^{-iZt} = \cos(t)I - i \sin(t) Z = \begin{pmatrix} e^{-it/2} & 0 \\ 0 & e^{it/2} \end{pmatrix}$$

$$R_Z(2t)$$

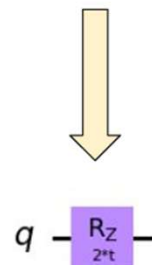
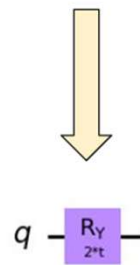
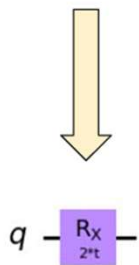


These are just **Rotational gates** on Bloch sphere!

$$e^{-iXt}$$

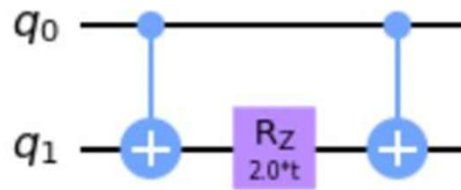
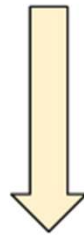
$$e^{-iYt}$$

$$e^{-iZt}$$



Multiple Qubit Simulation

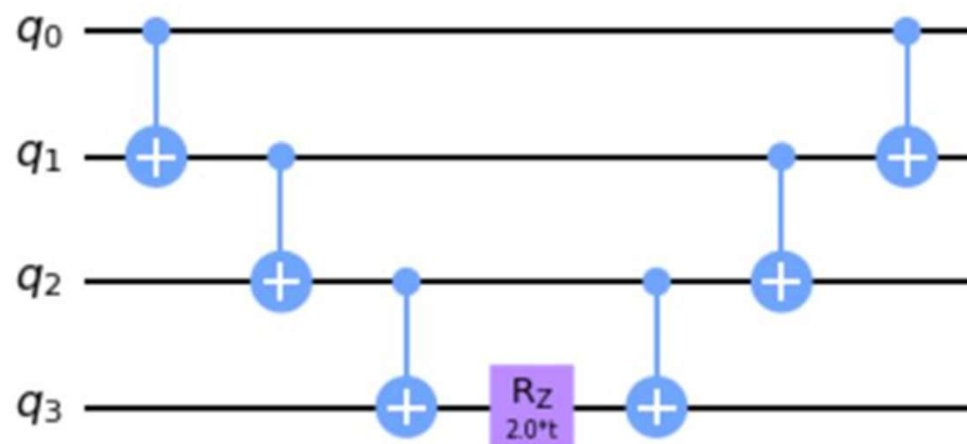
$$\exp(-iZ_0Z_1t) = \text{CNOT}_{0,1} \exp(-iZ_1t) \text{CNOT}_{0,1}$$



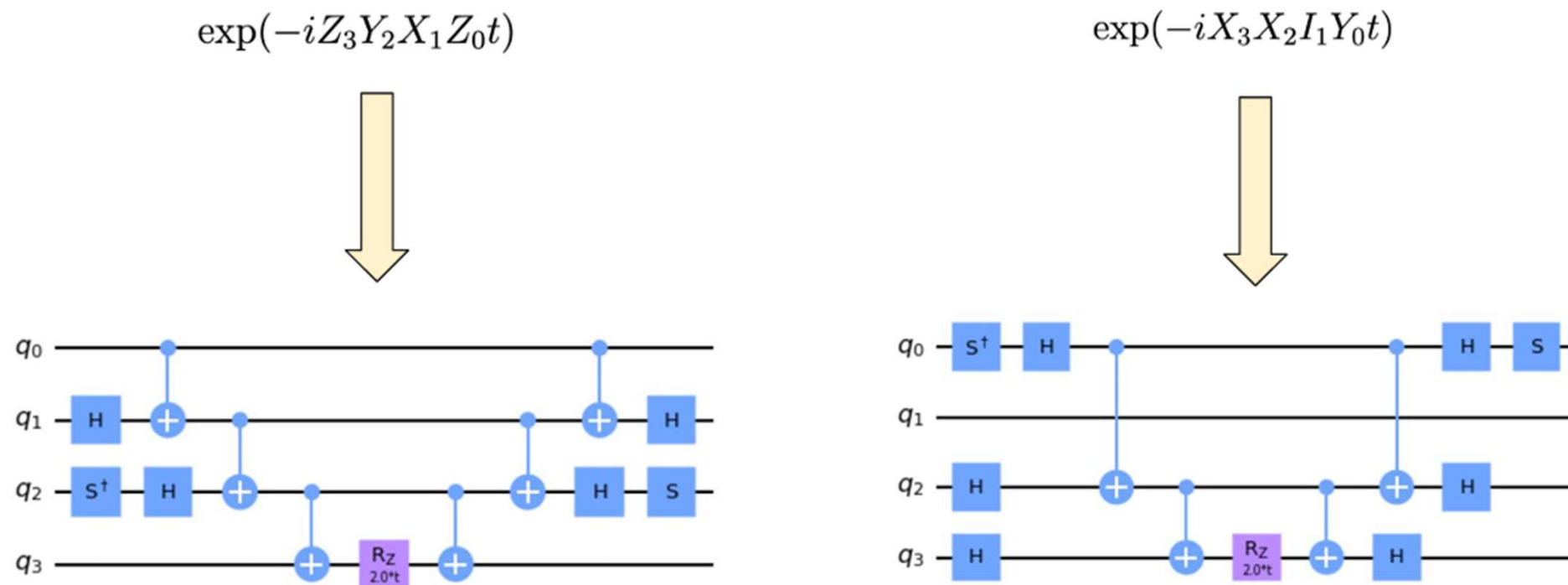
Multiple Qubit Simulation

The results can be generalized to $\exp(-i ZZ\dots Z t)$, where the circuit is **a tower of control gates**

For example, $\exp(-i ZZZZ t)$ becomes



Multiple Qubit Simulation



Pic. 50 Quantum Circuit Decomposition of Multi-Qubit Pauli Exponentials

$$X = HZH^\dagger, \quad Y = (SH)Z(SH)^\dagger$$

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$S = \sqrt{Z} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

Combinatorial Optimization: MaxKcut

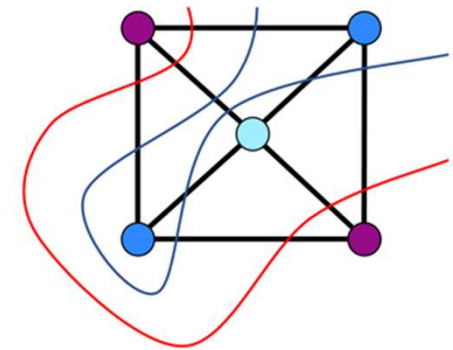
$$\max_{X \in \{1, \dots, k\}^n} C(X), C(X) = \sum_{(i,j) \in E} w_{ij} \begin{cases} 1, & \text{if } X_i \neq X_j \\ 0 & \text{otherwise} \end{cases}$$

Solving NP hard optimization problems.

- **Heuristic algorithms:** No polynomial run time guarantee; appear to perform well on some instances.
- **Approximate algorithms:** Efficient and provide provable guarantees. With high probability we get a solution X^* such that

$$\frac{C(X^*) - \min_x C(x)}{\max_x C(x) - \min_x C(x)} \geq \alpha,$$

where $0 < \alpha \leq 1$ is the approximation ratio.



k	2	3	4	5
α	.878567	.836008	.857487	.876610

k	6	7	8	9
α	.891543	.903259	.912664	.920367

The Problem Hamiltonian

For a given k we encode the information of a vertex belonging to one of the sets by $|i\rangle_{L_k}$, which requires

$$L_k := \lceil \log_2(k) \rceil$$

qubits.

$$H_p = \sum_{e \in E} W_e H_e, H_e = I - \widehat{H}_e, \quad U_p(\gamma) = e^{i\gamma H_p} = \prod_{e \in E} e^{i\gamma W_e \mathbb{I}} e^{-i\gamma W_e \widehat{H}_e}$$

$$\widehat{H}_e = \sum_{i,j=1}^{2^{L_k}} |ij\rangle \langle ij|$$

$i, j \sim_{clr} j$

Here, $\sim_{clr}$ is an equivalence relation $clr = \{\text{sets of equivalent colors states}\}$. With this, it is clear that $H_{i,j}$ has eigenvalue 0 for a state $|i\rangle|j\rangle$ if $i \tilde{c} \sim_{clr} j$ and +1 if not.

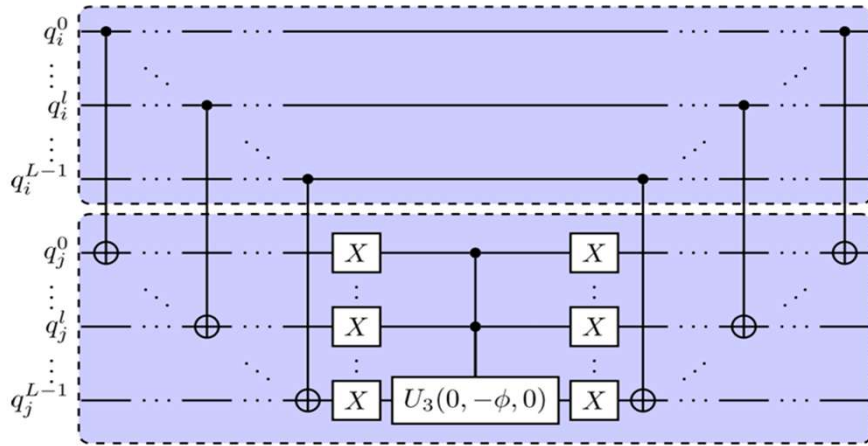
The Power of Two Case

When $k = 2^L$ the equivalence relation for the problem Hamiltonian becomes trivial because each color $C_i \rightarrow |bin(i)\rangle$ we assigned the related binary state

$$\widehat{H}_e = \sum_{i=0}^{k-1} |i\rangle \langle i| \otimes |i\rangle \langle i| = \sum_{i=0}^{k-1} |ii\rangle \langle ii| = CX_{A \rightarrow B} (I_A \otimes |0\rangle_B \langle 0|_B) CX_{A \rightarrow B}$$

$$CX_{A \rightarrow B} |q\rangle_A |q\rangle_B = |q_A^0\rangle \dots |q_A^{L-1}\rangle |q_B^0 \oplus q_A^0\rangle \dots |q_B^{L-1} \oplus q_A^{L-1}\rangle$$

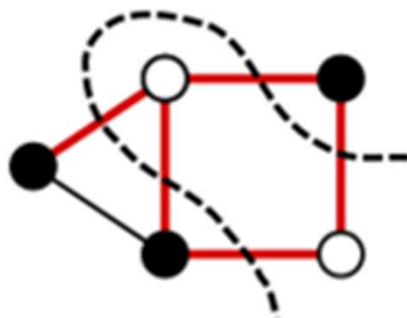
Where the $\oplus$ operation is modulo 2. This means that the state of the qubits belonging to j has zero entries if and only if all qubits have the same state as in the register i .



Pic. 51 Comparator Circuit for the Problem Hamiltonian

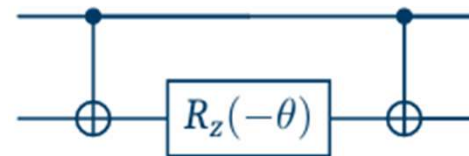
$$e^{-i\phi \widehat{H}_e} = CX_{A \rightarrow B} (I_A \otimes e^{-i\phi |0\rangle_B \langle 0|_B}) CX_{A \rightarrow B}$$

The Maxcut Case



$$\widehat{H}_e = |00\rangle\langle 00| + |11\rangle\langle 11| = \frac{\mathbb{I} + Z \otimes Z}{2} \Rightarrow H_{Maxcut} = \sum_{(i,j) \in E} W_{ij} \frac{1 - Z_i Z_j}{2}$$

$$e^{-i\theta Z \otimes Z} = \begin{pmatrix} e^{-i\theta/2} & 0 & 0 & 0 \\ 0 & e^{i\theta/2} & 0 & 0 \\ 0 & 0 & e^{i\theta/2} & 0 \\ 0 & 0 & 0 & e^{-i\theta/2} \end{pmatrix} =$$



The Degenerate Color Method

When $k \neq 2^{L_k}$ we can think to encode the k colors into all the computational basis states with some equivalence class degeneracy for some choices of colors. One possible choice is to assign $C_0 \rightarrow |0\rangle$, $C_1 \rightarrow |1\rangle$, ..., $C_{k-2} \rightarrow |k-2\rangle$, $C_{k-1} \rightarrow \bigcup_{i=k}^{2^{L_k}-1} \{|i\rangle\}$.

$$\widehat{H}_e = \sum_{i,j=1}^{2^{L_k}} |i\rangle\langle i| \otimes |j\rangle\langle j| = \sum_{i,j=1}^{2^{L_k}} |ij\rangle\langle ij|$$

We should find a way to construct the exponential $U_p(\gamma) = e^{i\gamma \cdot p}$ using the minimum amount of resources for the specific choices of k and the equivalence class clr .

$$C_0 \rightarrow |00\rangle, C_1 \rightarrow |01\rangle, C_2 \rightarrow |10\rangle, |11\rangle$$

$$\widehat{H}_e = |0000\rangle\langle 0000| + |0101\rangle\langle 0101| + |1010\rangle\langle 1010| + |1111\rangle\langle 1111| + |1110\rangle\langle 1110| + |1011\rangle\langle 1011|$$

Realizing exponential of diagonal Projecting operators

We want to realize the exponential of a *diagonal square* binary matrix Λ , i.e.,

$$\Lambda_{i,j} \in \begin{cases} \{0, 1\}, & \text{if } i = j \\ \{0\}, & \text{otherwise,} \end{cases} \quad \forall 0 \leq i, j \leq 2^n - 1$$

with quantum gates. Let's start with the case of where the number of ones elements in Λ are m , a power of two. There exists a permutation P_π , such that

$\Lambda = P_\pi(|0\rangle_A \langle 0|_A \otimes I_B) P_\pi^{-1}$ where $|B| = \log(m)$ and $|A| = n - |B|$. From this it follows that

$$\begin{aligned} e^{i\theta\Lambda} &= \left(\sum_{l=0}^{\infty} \frac{(i\theta)^l}{l!} \Lambda^l \right) = I + \left(\sum_{l=0}^{\infty} \frac{(i\theta)^l}{l!} \Lambda^l \right) = I + (e^{i\theta} - 1)\Lambda \\ &= P_\pi \left((I_A + (e^{i\theta} - 1)|0\rangle_A \langle 0|_A) \otimes I_B \right) P_\pi^{-1} = P_\pi (e^{i\theta|0\rangle_A \langle 0|_A} \otimes I_B) P_\pi^{-1} \end{aligned}$$

Realizing exponential of diagonal Projecting operators

In general, we can rewrite m as $m = 2^{q_1} + 2^{q_2} + \dots + 2^{q_p}$ and consequently, divide Λ into a sum of diagonal matrices with corresponding many non-zero elements, i.e.,

$$\Lambda = \Lambda_{q_1} + \Lambda_{q_2} + \dots + \Lambda_{q_p}$$

For all matrices Λ_{q_i} in the above sum the number of non-zero elements is a power of two, then

$$e^{i\theta\Lambda} = \prod_{j=1}^P e^{-i\theta\Lambda_j} = \prod_{j=1}^P P_{\pi_j} \left(I_{A_j} \otimes D_{Z_j} \right) P_{\pi_j}^{-1} = \widehat{P_{\pi_0}} \prod_{j=1}^P \widehat{D_j} \widehat{P_{\pi_j}}$$

Each permutation P_{π} can be in principle implemented using only gates from the set $\{X, CX, CCX\}$.

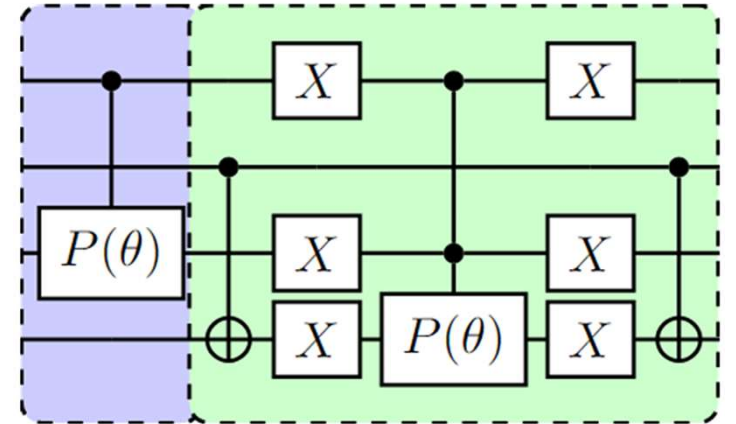
The K=3 case

In the edge Hamiltonian $|0000\rangle, |0101\rangle, |1010\rangle, |1111\rangle, |1011\rangle, |1110\rangle$ are the +1 eigenstates so the number of ones in $\widehat{H}_e$ is $m = 6 = 2^2 + 2^1$. Dividing $\widehat{H}_e = H_1 + H_2$, where H_1 represents the first two colors and H_2 the remaining ones.

$$H_1 = \sum_{(i,j) \in \text{clr}_{\leq 3}^3 | i, j \leq 1} |ij\rangle\langle ij| = CX_{2,4}(|0\rangle\langle 0| \otimes I \otimes |0\rangle\langle 0| \otimes |0\rangle\langle 0|)CX_{2,4},$$

$$H_2 = \sum_{(i,j) \in \text{clr}_{\leq 3}^3 | i, j \geq 2} |ij\rangle\langle ij| = X_1X_3(|0\rangle\langle 0| \otimes I \otimes |0\rangle\langle 0| \otimes I)X_1X_3,$$

Where $CX_{a,b}$ denotes a CX gate with control index a and target index b .



The Constrained to a Subspace method

When k is not a power of two instead of using equivalence color classes we can define the feasible set B and constraint the QAOA ansatz to explore only this set. We then need to initialize a feasible initial state and define a constraint Mixer.

For each vertex the feasible set of color states can be defined as:

$$B_k = \{|i\rangle | 0 \leq i \leq k - 1\}$$

Since the feasible bit-strings admit a Cartesian product structure (the vertices are independent), the feasible subspace becomes a tensor product of the form

$$B = B_k^1 \otimes \cdots \otimes B_k^{|V|}$$

We observe that

$$H_P^{bin}(2^{L_k})|_{span(B)} = H_P^{bin}(k),$$

Using this approach, we shift the complexity from the Phase or Cost Hamiltonian to the Mixer Hamiltonian.

Constrained QAOA

The solutions constrained to a feasible subspace $\text{span}(B) \subset \mathcal{H} = (\mathbb{C}^2)^{\otimes n}$

$$B_k = \{|Z_j\rangle, 1 \leq j \leq J, \quad Z_j \in \{0, 1\}^n\}.$$

Definition valid mixer

- Preserve the feasible subspace

$$U_M(\beta)|V\rangle \in \text{span}(B), \quad \forall |V\rangle \in \text{span}(B), \quad \forall \beta \in \mathbb{R},$$

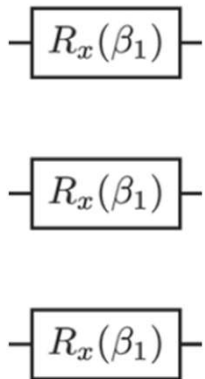
- Provide transitions between all pairs of feasible states, i.e., for each pair of computational basis states $|x\rangle, |y\rangle \in B$ there exist $\beta^* \in \mathbb{R}$ and $r \in \mathbb{N} \cup \{0\}$, such that

$$|\langle x | \underbrace{U_M(\beta^*) \dots U_M(\beta^*)}_{r \text{ times}} | y \rangle| > 0.$$

Example of Valid Mixers

Unconstrained case: X mixer

$$U_X(\beta) = \prod_i R_{X_i}(\beta) = \prod_i (\cos(\beta)\mathbb{I} + i \sin(\beta)X_i)$$

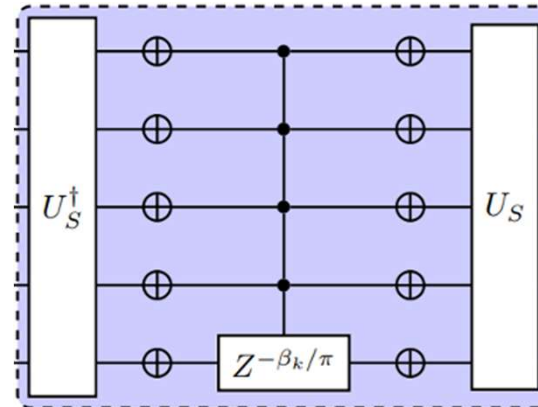


$$U_X\left(\frac{\pi}{2}\right) = \frac{1}{2\sqrt{2}} (\mathbb{I} + i(X_1 + X_2 + X_3) - (X_1X_2 + X_2X_3 + X_1X_3) - iX_1X_2X_3)$$

Constrained case: Grover mixer

$$|F\rangle = \frac{1}{\sqrt{|B|}} \sum_{i \in B} |i\rangle = U_S |0\rangle \Rightarrow$$

$$U_{Grover}(\beta) = e^{i\beta|F\rangle\langle F|} = U_S e^{i\beta|0\rangle\langle 0|} U_S^\dagger$$



$$(|F\rangle\langle F|)^2 = |F\rangle\langle F| \Rightarrow U_{Grover}(\beta) = \sum_i \frac{(i\beta)^n}{n!} (|F\rangle\langle F|)^n = \mathbb{I} + (e^{i\beta} - 1)|F\rangle\langle F|$$

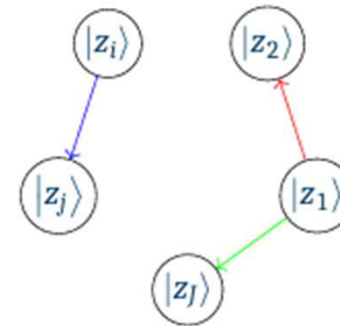
Valid Mixers as Connected Graphs

We consider mixers of the following form

$$U_M(\beta) = e^{-i\beta H_M}, \quad H_M = \sum_{j < k} (T)_{j,k} H_{Z_j \leftrightarrow Z_k}, \quad |Z_j\rangle, |Z_k\rangle \in B$$

$$H_{Z_j \leftrightarrow Z_k} = |Z_j\rangle\langle Z_k| + |Z_k\rangle\langle Z_j| = X_{Z_k \leftrightarrow Z_j}(|Z_j\rangle\langle Z_j| + |Z_k\rangle\langle Z_k|) = X_{Z_k \leftrightarrow Z_j} \Pi_{Z_k, Z_j}.$$

$$T = \begin{matrix} & |z_1\rangle & |z_2\rangle & \cdots & |z_i\rangle & |z_j\rangle & \cdots & |z_J\rangle \\ \begin{matrix} |z_1\rangle \\ |z_2\rangle \\ \vdots \\ |z_i\rangle \\ |z_j\rangle \\ \vdots \\ |z_J\rangle \end{matrix} & \begin{pmatrix} & & & & & & \\ & \textcolor{red}{1} & & & & & \\ & \textcolor{red}{1} & & & & & \\ & & & & & & \\ & & & & & \textcolor{blue}{1} & \\ & & & & \textcolor{blue}{1} & & \\ & & & & & & \\ & & & & & & \\ \textcolor{green}{1} & & & & & & \end{pmatrix} \end{matrix}$$

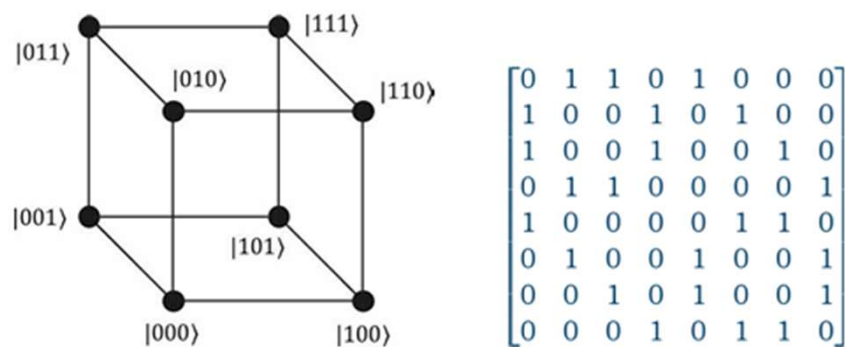


Valid mixer

The mixer Hamiltonian H_M is valid if the graph G_T of the adjacency matrix T is undirected and connected.

Examples of Graphs and Mixers

X mixer as Hypercube Graph

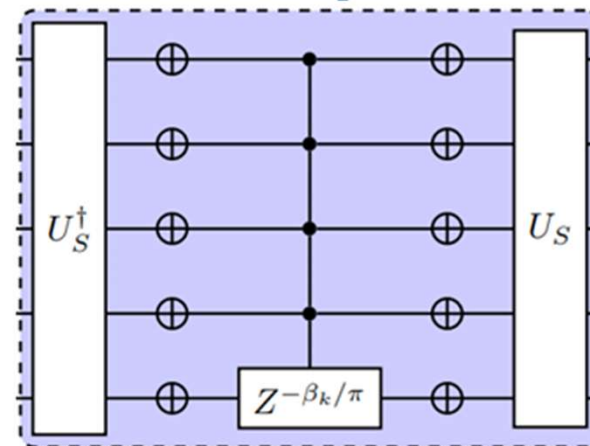
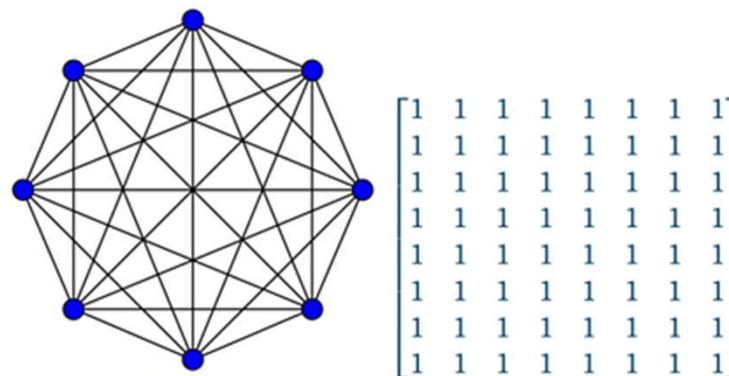


$$R_x(\beta_1)$$

$$R_x(\beta_1)$$

$$R_x(\beta_1)$$

Grover mixer as Complete Graph



Initial State Preparation

Given the independence of the different color constraints for each vertex of the problem the initial state can be prepared as the uniform superposition of all feasible basis state as follows:

$$|F\rangle^{\otimes |V|} = U_S \otimes U_S \dots \otimes U_S |0\rangle, \text{ where } |F\rangle = U_S |0\rangle = \frac{1}{\sqrt{k}} \sum_{i=0}^{k-1} |i\rangle$$

Where we prepare independently on each vertex the equal superposition of all feasible colors due to the property of the tensor product:

$$\left(\sum_{i \in B_k} a_i |i\rangle \right) \otimes \left(\sum_{j \in B_k} b_j |j\rangle \right) = \sum_{i \in B_k, j \in B_k} a_i b_j |ij\rangle = \sum_{m \in B_k \otimes B_k} c_m |m\rangle$$

Grover Mixer construction

Now we need to define a valid mixer for the feasible subspace B. The first possibility is to use directly the initial state preparation circuit $U_S \otimes \cdots \otimes U_S$ to implement the Grover mixer

$$U_{Grover}(\beta) = (U_S \otimes \cdots \otimes U_S) e^{i\beta(|0\rangle\langle 0|)^{\otimes |V|}} (U_S^\dagger \otimes \cdots \otimes U_S^\dagger)$$

The total cost of this mixer is $2|V|$ times the cost of implementing the initial state for a vertex and the cost of implementing the $|V| L_k - 1$ control phase operator. Alternative we can take advantage from the structure of B and define the Grover Mixer independently for each vertex.

$$U_M(\beta) = U_S e^{i\beta|0\rangle\langle 0|} U_S^\dagger \otimes \cdots \otimes U_S e^{i\beta|0\rangle\langle 0|} U_S^\dagger$$

Grover Mixer construction

This Grover mixer can be obtained by exponentiation of the following Hamiltonian Mixer

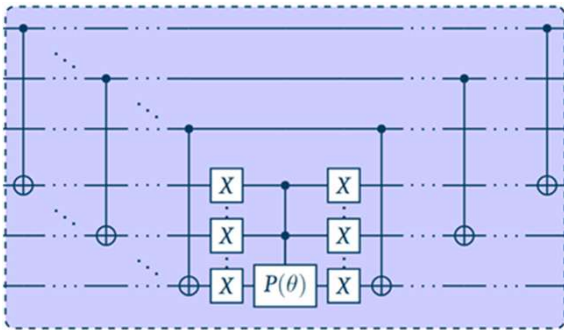
$$H_M = G \square^{|V|} = G \square G \dots \square G, \quad G = |F\rangle\langle F|$$

where the Cartesian or Box product is defined as

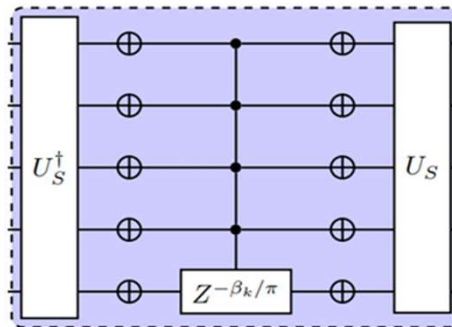
$$G \square H = G \otimes \mathbb{I} + \mathbb{I} \otimes H$$

If G and H are connected graphs, then $G \square H$ is connected too and the mixer is valid.

**Cost Circuit for an edge of the
Maxcut problem**



**Grover Mixer for a $16 < k \leq 32$
MaxKCut vertex**



**Initial state for a $16 < k \leq 32$
MaxKCut vertex**

